ASSIGNMENT-3.2

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BATCH-16

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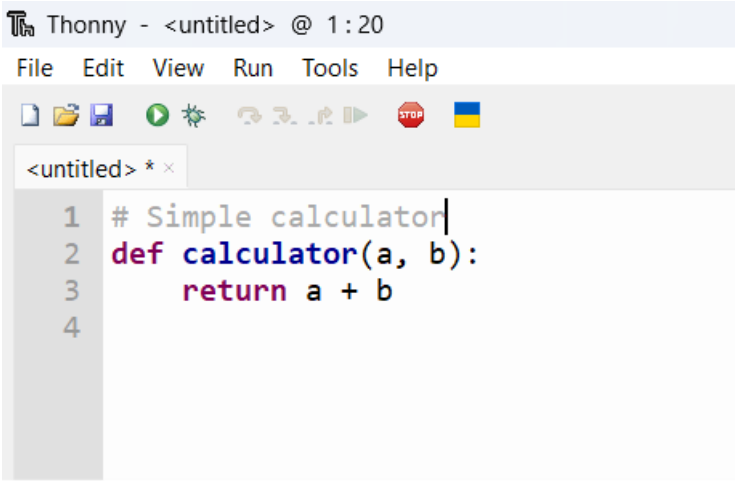
Task 1: Progressive Prompting for Calculator Design

Initial AI Output (Basic Calculator)



Improved AI Output (With Comments & Usage Examples):-

Thonny - <untitled> @ 27 : 1

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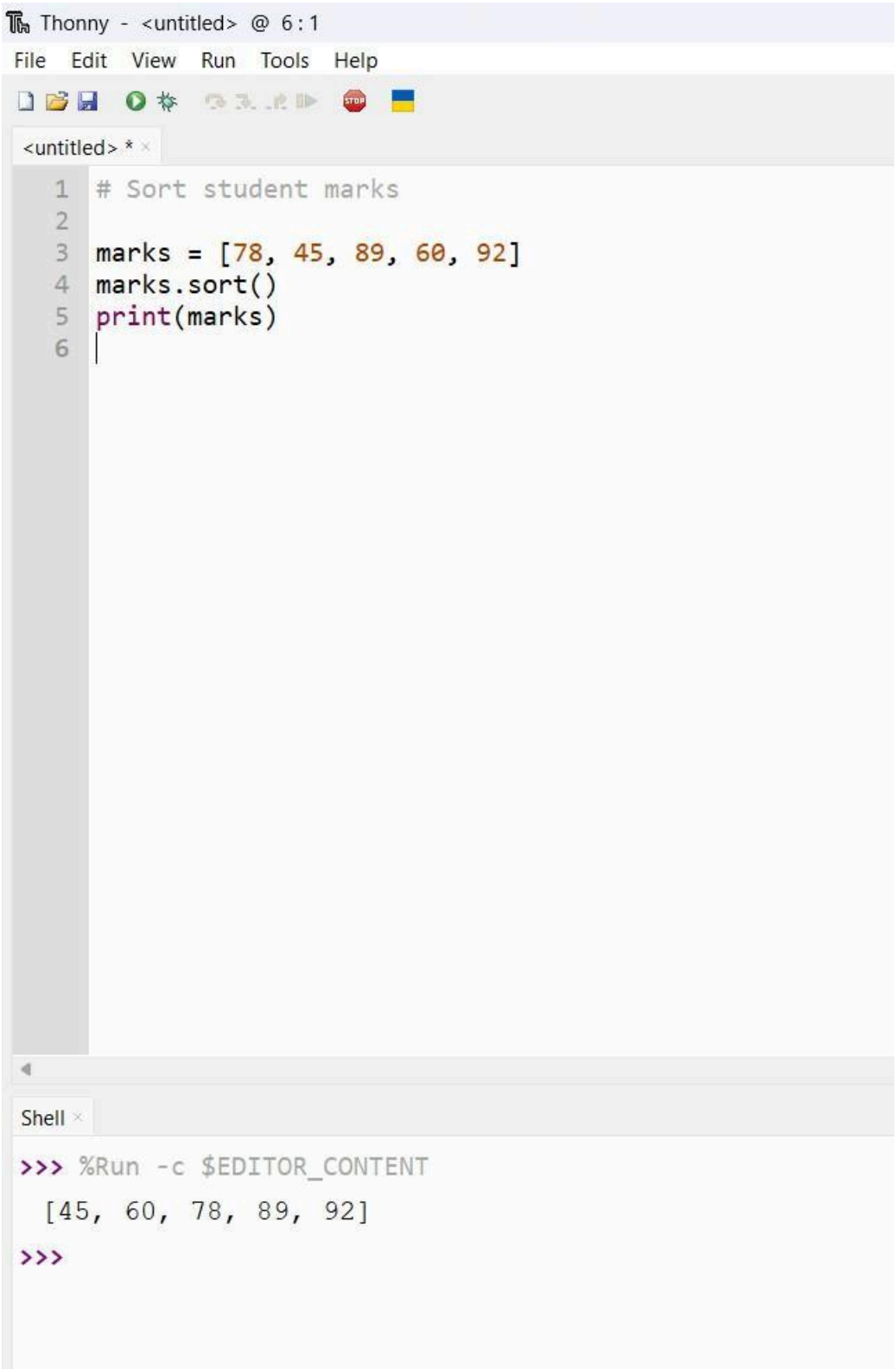
```python
# Calculator program using functions
# Supports addition, subtraction, multiplication, and division

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Division by zero not allowed"
    return a / b


# Example usage
x = int(input("Enter first number: "))
y = int(input("Enter second number: "))

print("Addition:", add(x, y))
print("Subtraction:", subtract(x, y))
print("Multiplication:", multiply(x, y))
print("Division:", divide(x, y))
```

Shell

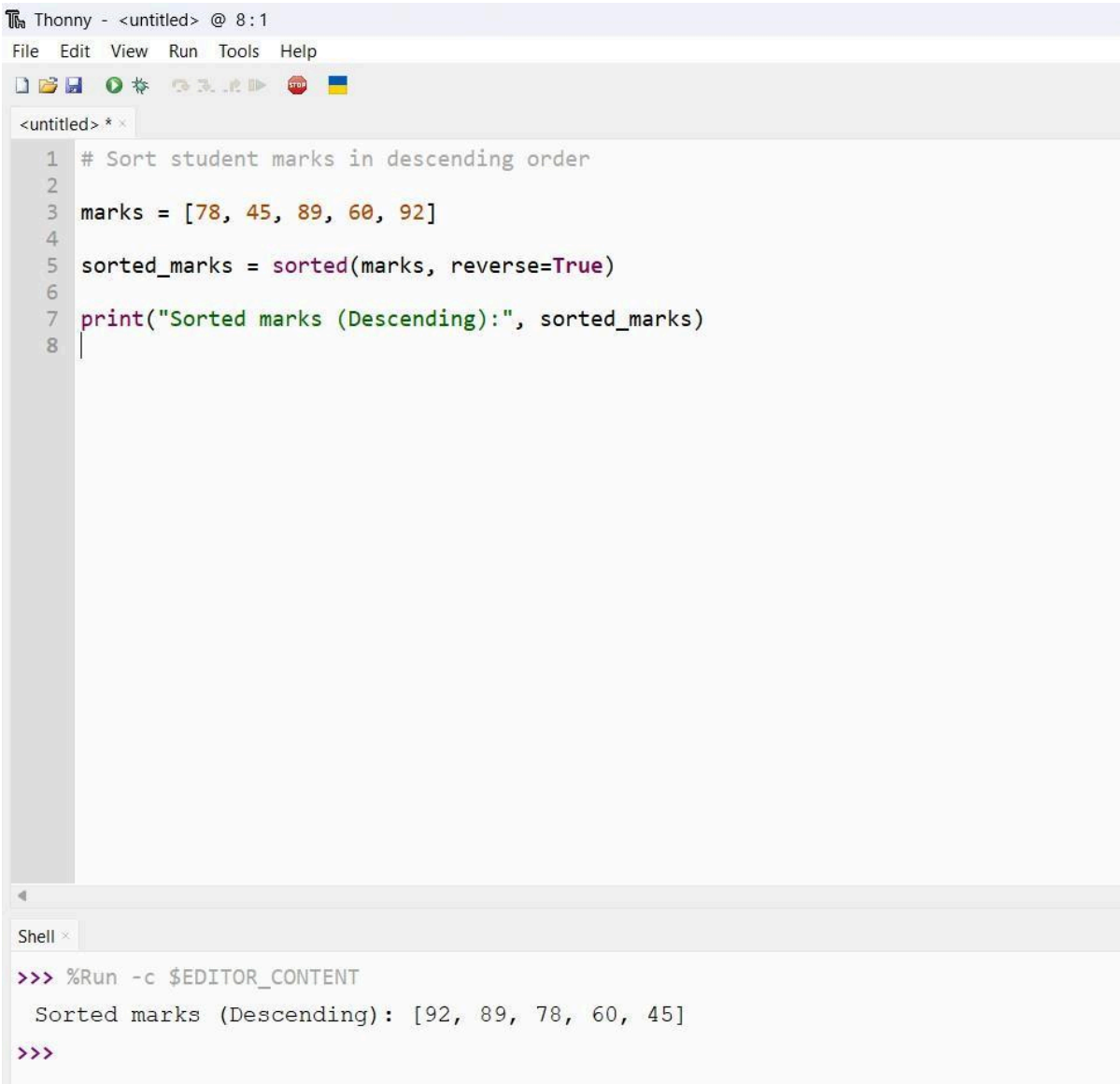
```
Enter first number: 4
Enter second number: 9
Addition: 13
Subtraction: -5
Multiplication: 36
Division: 0.4444444444444444
>>>
```

Task 2: Refining Prompts for Sorting Logic

Initial AI Output (Vague Prompt):-



Refined AI Output (Clear Prompt with Order & Constraints):-



Task 3: Few-Shot Prompting for Prime Number Validation

AI-Generated Prime Checking Function:-

Thonny - <untitled> @ 19 : 1

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```python
# Function to check whether a number is prime

def is_prime(num):
    if num <= 1:
        return False

    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            return False

    return True


# Test cases
print(is_prime(2))
print(is_prime(7))
print(is_prime(10))
print(is_prime(1))
```

Shell

```
>>> %Run -c $EDITOR_CONTENT

 True
 True
 False
 False

>>>
```

Task 4: Prompt-Guided UI Design for Student Grading System:-

Thonny - <untitled> @ 27 : 1

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```python
# Student grading system

name = input("Enter student name: ")
marks1 = int(input("Enter marks in Subject 1: "))
marks2 = int(input("Enter marks in Subject 2: "))
marks3 = int(input("Enter marks in Subject 3: "))

total = marks1 + marks2 + marks3
percentage = total / 3

if percentage >= 90:
    grade = "A"
elif percentage >= 75:
    grade = "B"
elif percentage >= 60:
    grade = "C"
elif percentage >= 40:
    grade = "D"
else:
    grade = "Fail"

print("\n--- Student Report ---")
print("Name:", name)
print("Total Marks:", total)
print("Percentage:", percentage)
print("Grade:", grade)
```

Shell ×

```
>>> %Run -c $EDITOR_CONTENT

 Enter student name: ROHAN
 Enter marks in Subject 1: 87
 Enter marks in Subject 2: 96
 Enter marks in Subject 3: 98

 --- Student Report ---
 Name: ROHAN
 Total Marks: 281
 Percentage: 93.66666666666667
 Grade: A

>>> |
```

Task 5: Prompt Specificity in Unit Conversion Functions:-

Thonny - <untitled> @ 13 : 1

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```python
# Unit conversion functions

def km_to_miles(km):
    return km * 0.621371

def miles_to_km(miles):
    return miles / 0.621371


# Test cases
print("10 km =", km_to_miles(10), "miles")
print("5 miles =", miles_to_km(5), "km")
```

Shell

```
>>> %Run -c $EDITOR_CONTENT

  10 km = 6.21371 miles
  5 miles = 8.046722489462816 km

>>>
```